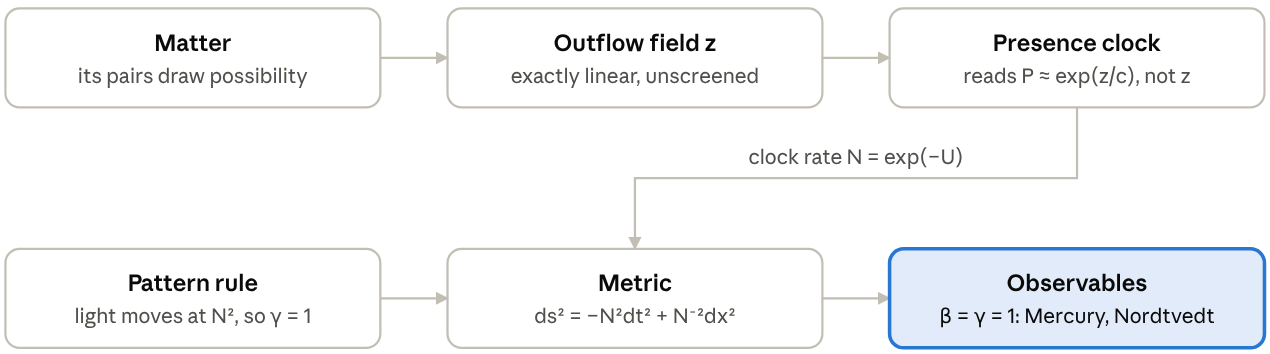
Gravity from Counting Events

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In Event Density's commitment ontology, time at a place is the count of events completing there, and matter slows that count around it. On ED's own event-layer simulations, the counting rules now give the exponential metric, with β = γ = 1, to the precision the runs reach. The event layer runs one way, and a clock's record averaged over time loses most of that order: time-symmetric laws as averages of one-way events. What fails is the field's dynamics: the well is carried by diffusion, so it cannot follow moving matter, and at the adopted scales it cannot form beyond about 0.3 μm. The static results in the table below are steady states that only a faster field rule (such as a deficit carried as a wave) would reach at physical distances.

How it works

Matter's draw sets a linear field; clocks read its exponential



from matter's draw to the metric · 6 steps

Matter's commitments draw possibility from their places. Conservation makes the outflow an exactly linear field of that draw, with no screening. A clock counts the possibility present, which grows exponentially with the outflow, so its rate is N = e^(−U) and two wells multiply. Light hops on ticks and needs free capacity, which sets the spatial part of the metric.

The rules

Space is a lattice of places. Each holds an integer amount of possibility P, and nothing in the vacuum commits. Each tick, every unit hops to a random neighbour with a chance chosen so that a place sends out

$$f(P) = c \ln\left(1 + \frac{P}{p_s}\right), \qquad c = \frac{0.99}{\ln(1 + 1/p_s)}$$

The outflow is linear in P below a crossover p_s and compares ratios above it. Matter is reserved pairs of neighbouring places. Each commitment takes one unit from each of its places, holds, and renews at once. Released units return uniformly. A commitment is also a clock: its hold ends when the possibility present at its places, summed over ticks, reaches a fixed amount.

$$\sum_{\text{ticks}} (P_a + P_b) \ \ge\ H = 2\hbar P_0$$

So a clock runs at a rate proportional to the possibility around it. Two results follow. First, conservation makes the mean outflow z = ⟨f(P)⟩ an exactly linear, unscreened field of matter's draw s:

$$\tfrac16 \sum_{y \sim x} z_y - z_x = s_x - \bar s$$

Second, the clock reads P, not z. In local equilibrium P = R(z), so the clock rate is N = R(z)/R(z_0) and

$$\beta = 1 + \frac{y''}{2\,y'^2}, \qquad y = \ln R(z)$$

When P ≫ p_s, ln P is linear in z, so N = e^(−U) and β → 1, with β − 1 ≈ −0.45 p_s/P_0. Light and other patterns of possibility hop on their place's ticks and need free capacity there, which fixes the spatial part:

$$ds^2 = -N^2\,dt^2 + N^{-2}\,d\mathbf{x}^2, \qquad \frac{1}{N^4}\,\partial_t^2 \phi = \nabla^2 \phi - \frac{m^2}{N^2}\,\phi$$

What the simulations show

Every check below was pre-registered: targets from measured physics, pass marks and kill rules hashed before the run.

Check	Target	Result on the event layer	Runs
Wells around matter	unscreened	outflow deficit matches its sinks with no free amplitude (1.00 ± 0.01)	14, 22
Clocks of different make	same redshift (null tests ~10^{-6})	gap between two designs falls as p_s^(−0.85): 0.056, 0.018, 0.005 at p_s = 1, 4, 16	22, 24
Free fall	every mass alike	patterns fall at 0.98–0.99 of the clock-set rate, masses alike to 0.6%; in the current rules' well, 0.986 with masses alike to 0.5%	16
Composite matter (knots)	falls like everything else	at its own lattice weight/inertia, 0.94–0.98, shortfall shrinking with the grid	18b
Light bending, γ	1 (Cassini)	2.010 × the clock-only bending (γ = 1 is put in by the pattern rule)	17
Perihelion, β	1 (Mercury, 42.98 ± 0.04″/century)	0.9996 at $P_0/p_s = 64$, within 0.0005 of 1 up to 4096	22, 23, 25
Two wells together	multiply (β = 1), not add	fit fraction f = 1.12 ± 0.19 over two seeds (1 = multiply, 0 = add)	26

Check	Target	Result on the event layer	Runs
Earth–Moon (Nordtvedt) test	$0 \pm 4.4\times10^{-4}$ (lunar ranging)	β part: -4.2×10^{-4} at $p_s = 16$, $P_0/p_s = 4096$. Rest mass in a potential pulls by its local clock rate, which is GR′s (Tolman′s) weight, so $\zeta_2 = 2(\beta - 1)$; the binary-pulsar bound on ζ_2 needs $P_0/p_s \gtrsim 2.3\times10^{4}$. Under the rules adopted 19 Oct, $\zeta_1 = \xi = \alpha_2 = 0$ follow from the rules and $\alpha_1 = \alpha_3 = 0$ from one tracked input (Lorentz invariance), so $\eta = (10/3)(\beta - 1) \approx -6.5\times10^{-5}$ at $P_0/p_s = 2.3\times10^{4}$, inside the bound.	25, 30, 34

Check	Target	Result on the event layer	Runs
Time′s arrow	laws whose averages carry no time order	the event layer runs one way (even the vacuum: 2.7×10^{-3} per place per tick); a clock′s record averaged over T ticks keeps an asymmetry that falls about as 1/T (0.034 → 0.0013 at 1024, reduced but not zero); over long windows the count becomes the sum of what it read; summing clocks does not reduce it. Under the adopted rules (Run 35) the vacuum is two-way (in detailed balance, derived), the record′s long tail vanishes, and averaging still removes the order; the arrow lives in commitment, the inflow to matter and clocks. Run 29′s long tail was carried by the medium′s ambient fluctuations, not each clock′s own hole (Run 37).	29, 35, 37

Check	Target	Result on the event layer	Runs
Moving bodies (field dynamics)	a body's field follows it; changes travel at c	engine as built (Run 31) fails: the well is carried by diffusion. 4% formed at 10 places 1,000 ticks after switch-on (light speed: 100%); a moving ball trails its well (fore-aft asymmetry 0.67 and 0.32, where a co-moving field gives 0). Static wells hold only as steady states, which diffusion at the adopted scales builds no further than about 0.3 µm. Adopted 19 Oct: the field is fixed each tick. On the engine (Run 35) a moving ball keeps its well and draw (asymmetry −0.002, draw 0.999 of static) at speeds up to 0.25 places per tick	31, 35

Check	Target	Result on the event layer	Runs
Horizons	GR: black holes have horizons and radiate at $T = \kappa/2\pi$	none from a static body: 16× the commitments gives 1.8× the draw, and the clock rate stays above zero outside matter; the idealised zero-clock surface has $\kappa = 0$, so no Hawking radiation. Under option A (Kerr exterior, a tracked input) the surface has GR's κ, so ED's radiation papers can apply, if the committing core is assumed to act on quantum fields as a horizon would: standard Hawking radiation from the surface (nothing released from the core), with no ED-specific signal. Allen ruled (19 Oct): black holes evaporate, assuming the core acts as a horizon for quantum fields and its pull drops while its record stays. The endpoint (a Planck-mass remnant, or full evaporation, which Allen leans toward) is open	32; papers 039–052 (reading)

Check	Target	Result on the event layer	Runs
Frame dragging; preferred frame (α_1, α_2)	$\lvert\alpha_1\rvert < 4\times10^{-5}$ (pulsar PSR J1738+0333), $\lvert\alpha_2\rvert < 2\times10^{-9}$. Frame dragging is $1 + \alpha_1/8$ of GR's, so GP-B's 37.2 ± 7.2 mas/yr tests the same number, $10^4\times$ more loosely	no current field: $\alpha_1 = -8$, $\alpha_2 = -1$ (excluded); the engine as built has no frame dragging either way. If patterns ride possibility's flow, $\alpha_2 = 0$ follows from conservation. α_1 needs a momentum channel of strength 4 (input). No derived rule passes. Adopted 19 Oct with one tracked input, Lorentz invariance at first post-Newtonian order (as for MICROSCOPE); the tick is not physical time. That passes α_1, α_3, frame dragging and the whole Earth–Moon test. On the engine a pattern co-rotates with a rotating ring at the current field's speed (Run 38, a construction check)	34 (calculation)

Check	Target	Result on the event layer	Runs
Energy pulls (a bound pattern)	a bound system pulls by its total energy, binding counted once (Al vs Fe alike to 4×10^{-12})	Run 36′s kill rule fired (+13% with compact matter held on fixed places). With bodies that keep their proper size, in an unbounded medium: at weak field the pull is ħω with binding counted once (−0.05 of the binding; energy alone misses by two bindings, −2.0); at strong field it misses by about one binding, a second-order departure from GR. Inside a body (atoms, nuclei) at weak field it is counted once only if the body′s internal stress pulls too, with weight 1 per direction (Run 42: −0.09 of the binding with stress, −0.83 without); adopted 19 Oct as part of the Lorentz input (a tracked input). On a grid twice as fine the pass holds and what remains is second order (Run 44), negligible for ordinary bodies but possibly large for neutron stars,	36, 40, 42, 44

Check	Target	Result on the event layer	Runs
		which pulsar tests bound tightly (not yet computed)	
Ringdown with spin	GW250114 (spin 0.68): frequency +2 ± 2%, damping time −1 +10/−9% of Kerr (90%)	fails at a = 0.68 under both rotating exteriors of the adopted metric (scalar eikonal): the engine′s own rule has no prograde light ring outside its throat and damps 2–3× too fast; the Newman–Janis completion rings 17% low. Allen chose (19 Oct): outside compact objects the field is GR′s (Kerr′s) as a tracked input, with ED′s no-singularity core instead of a horizon; so the ringdown and shadows match by construction, and the core′s own tests follow (next row)	39, 41 (calculations)

Check	Target	Result on the event layer	Runs
ED′s core (option A)	no echoes 50–350 ms after mergers (GWTC-3); no surface glow from Sgr A*; spinning objects without a horizon go unstable unless they absorb	viable if the core absorbs at least a few tenths of a percent of what reaches it (≥ 0.3% quenches the instability at GW250114′s spin; the glow bound at ED′s scale needs only ≳ 6×10^{-7}). Allen ruled (19 Oct): what reaches the core commits, so the core absorbs everything and passes every bound. From outside it looks exactly like GR′s black hole (no echoes: a partly reflecting core would have sent fainter repeats of the ringdown, at least 29 ms apart); inside there is no singularity, and what falls in is kept as record	43 (calculation)

Check	Target	Result on the event layer	Runs
Neutron stars (pull and fall)	a neutron star falls like white dwarfs to 2×10^{-6} (pulsar triple PSR J0337+1715)	a body at neutron-star strength: its own binding pulls once (first order) and falls within what the engine resolves ($\leqslant 2\times10^{-3}$, shrinking toward the physical medium and with the grid). A pattern bound inside it falls wrongly by about one binding, because the engine codes massive patterns as flat-space waves with a redshifted mass: fails as coded. Read as the full pattern rule (the wave equation in $-N^2dt^2 + N^{-2}dx^2$), the first-order failure is gone (Run 46): weak-field binding still counts once, and with a converged lookup table (Runs 46b, 46c) a bound pattern falls like its energy at neutron-star strength to a few $\times10^{-4}$ by its frequency, and to about 2×10^{-3} by its far field (whose limit on a finer grid is not determined); the body's own fall passes as registered. Nothing was falsified, and the line is closed at the	45, 46, 46b, 46c, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56

Check	Target	Result on the event layer	Runs
		engine′s resolution. Run 48 made the fall exact: in a scale-free medium an outside potential is exactly a finer grid (to 10^{-16}), so bound matter falls like its energy in the continuum (numerically to a few $\times 10^{-5}$); the pulsar triple then bounds how far ED′s medium departs from scale-free across a star. A medium with a controlled departure from scale-free shifts the fall linearly at the predicted size (Run 51). The pull there was 4% short of a bound pattern′s energy with the old body model; with matter in exact hydrostatic balance the engine′s statics come from one action, its discrete first law holds to 5×10^{-8}, and the far field rises by $\hbar\omega$ (Runs 53, 56, registered: the surface-averaged continuum miss is within about 1% of $\hbar\omega$, against −0.20 b with the old body model): the same first law as GR, not GR′s mass–	

Check	Target	Result on the event layer	Runs
		radius relation. In a medium with a scale, the variational draw $T(1 + g)$ keeps the first law exact; under it a body's equivalence-principle violation is $\Delta = M_{far}/E_H - 1 \approx \bar{g} - g_{bg}$, confirmed in two medium families and not additive (Runs 52, 54); Allen adopted that draw. ED's own medium has a scale ($g = (p_s - b)/P \approx p_s/P$), so with inertia = total energy (Allen's ruling) ED predicts $\Delta \approx C \cdot p_s/P_0 > 0$, with $C \approx 0.33$–0.49 for the pulsar; the triple needs $P_0/p_s \gtrsim 1.4$–2.9×10^5 (Run 55). First order: the pulsar bound needs $P_0/p_s \gtrsim 10^5$	

The engine adopted before these rules gave $\beta = 0.47$, which puts Mercury at 50.5″ per century. That failure is what forced the current rules.

What is put in, and what is open

Some of the agreement is built in, not derived:

- **$\gamma = 1$** comes from the pattern rule (light needs ticks and free capacity), chosen to give it.
- **Twist charged on tick-relation loops at equal price** makes knots fall universally. It is Lorentz invariance put in by hand.

- **The ratio medium and the presence clock** were designed to give β = 1. Their achievement is passing the independent checks, not producing β = 1 unasked.
- **The scales are bounds, not predictions:** $p_s \gtrsim 4\times10^5$ so that clocks agree to 10^{-6}, and $P_0 \gtrsim 2.3\times10^4$ $p_s \approx 10^{10}$ units per place for the binary-pulsar bound on ζ_2 (2000 p_s for Nordtvedt's β part). Both are extrapolated from the engine's reach. They say matter is light next to the medium, which is also what weak gravity needs.

Open:

- the field's dynamics (Run 31): the engine's well forms and moves by diffusion. At the adopted scales a well would spread about 0.3 μm in the age of the universe, and a body at 370 km/s would keep its well within 0.01 of a place. The fix needs a rule: a deficit carried as a wave (same static wells, changes at c), or places that move with matter
- the strong field: the metric has no horizon (a throat at areal radius e M). Its shadow is 4.6% larger than GR's, which EHT allows at 2σ. Its static ringdown frequency is about 4% below Kerr's, roughly 5σ from GW250114 if that carries over to spinning remnants and ED's own waves. ED's Hawking-radiation papers assume a horizon this metric lacks; a static body can never make one, because its draw throttles itself, and even the idealised surface where possibility runs out is at zero temperature (Run 32) (Run 28)
- a preferred frame (the lattice has a universal tick), and how moving matter and stress source gravity. Rest mass in a potential already pulls with GR's weight (Run 30); moving matter must draw $1 + 2v^2$ times its ticking rate, and the engine has no rule for moving matter yet
- knots built from the engine's own commitments
- patterns falling close in to deep wells, where the measuring method itself fails (Run 27)